

POWER BI CAPSTONE PROJECT REPORT

YouTube Creator Engagement & Optimization Analysis

An End-to-End Business Intelligence Case Study on Global Video Analytics and Platform Benchmarking

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1. Executive Introduction & Project Overview

In the modern digital economy, data-driven insights form the cornerstone of scalable content strategy and digital marketing. As a Power BI developer, this project focuses on evaluating a comprehensive global YouTube dataset derived from the platform's API. The dataset encompasses varied critical operational pillars including viewer metrics, channel structural properties, global ranking distributions, and specific category clusters.

The core objective is to architect, transform, model, and visualize these structured indicators to empower creative stakeholders, corporate brand networks, and performance marketers. By discovering hidden audience dynamics, this dashboard provides strategic frameworks to optimize runtime formats, drive conversion metrics, and dramatically scale interaction ratios.

2. Problem Statement & Strategic Scope

The baseline operational landscape exhibits deep volatility across consumer interaction points. Channels frequently scale absolute subscriber counts without seeing equivalent baseline view frequency or engagement metrics. This analysis systematically answers foundational questions surrounding platform optimization:

- What structural factors isolate hyper-productive channels regarding total media volume?
- How do platform categories cluster with respect to absolute subscriber weights?
- What is the exact correlation profile between baseline visibility (view ranks) and cross-channel type execution?

3. Data Lifecycle: Extract, Transform, & Load (ETL)

To preserve integrity and guarantee analytical accuracy, raw dataset inputs underwent explicit conditioning inside the **Power Query Editor** prior to visualization compilation:

- **Null Value Mitigation:** Erroneous fields within historical metrics and rolling 30-day performance windows were identified and neutralized to eliminate mathematical bias during statistical aggregations.
- **Data Type Standardization:** High-weight parameters including Subscribers, Uploads, and Ranks were mapped strictly to integer whole numbers, while localized nomenclature fields were converted to structured Text schemas.
- **30-Day Velocity Alignment:** Outlier inputs and string formatting anomalies in rolling historical metrics were sanitized, ensuring precise execution of rolling average measures.

4. Technical Core Implementation (Tasks 1–15)

The project requirements dictate explicit analytics checkpoints. Below is the detailed technical breakdown of the implemented analytical matrix:

ID	Core Requirement Task	Power BI Desktop & DAX Implementation Matrix
1	Maximum Uploads Card	Standard single-value Card visual bound to Uploads using the MAX aggregation filter.
2	Minimum View Ranks Card	Standard KPI Card utilizing MIN aggregation mapping on platform view ranking parameters.
3	Average Views (Last 30 Days)	Card visual mapping the data-cleansed 30-day velocity metrics using the AVERAGE evaluation framework.
4	Line Chart (>2 Lakh Uploads)	X-axis: Title, Y-axis: Uploads (Max), filtered via Advanced Filter Pane where Uploads $\geq 200,000$.
5	Spider Map (Radar Chart)	Custom visualization integrating channel_type across categorical rings evaluated via channel_rank.
6	Packed Bubble Chart	Custom cluster model displaying category as group nodes weighted in size via Subscribers.
7	Viewed by Subscriber Metric	Calculated column isolating audience interaction density via structured division logic. (Formula detailed in Section 5).
8	Channel Interaction Table	Multi-dimensional matrix visual grouping channel_type alongside the aggregated engagement scores.
9	Total Uploads Measure	Card visual connected to an explicit DAX measure compiling complete platform operational scale.
10	Decomposition Tree	AI-driven exploration tree analyzing category, path-explained via hierarchical ranking parameters.
11	Blank Navigation Button	Interactive UI element configured via Action settings set to Page Navigation targeting secondary views.
12	Report Professional Formatting	Design optimization utilizing unified theme structures, typography hierarchies, and structured branding assets.

ID	Core Requirement Task	Power BI Desktop & DAX Implementation Matrix
13	Power BI Cloud Publication	Production deployment into cloud infrastructure via Power BI Service within a managed tenant workspace.
14	Executive Web Dashboard	Pinned higher-order tiles consolidated into a single cloud dashboard for real-time executive visibility.
15	Cross-Platform Export	Production compile from cloud workspace to portable presentation outputs (PDF/PPT layout format).

5. Advanced DAX Formula Architecture

To execute Task 7 and Task 9 according to performance enterprise standards, explicit Data Analysis Expressions (DAX) were engineered. Instead of basic arithmetic expressions, robust divide logic was used to safeguard visualizations against runtime mathematical division errors when encountering channels with zero or unrecorded subscriber counts:

Viewed by Subscriber = DIVIDE(SUM('Global YouTube Statistics'[video views]), SUM('Global YouTube Statistics'[subscribers]), 0)

For Task 9, to guarantee structural consistency independent of local filter cross-filtering constraints, an explicit aggregated measure was deployed:

Total Uploads = SUM('Global YouTube Statistics'[Uploads])

6. Strategic Insights & Analytical Key Takeaways

Insight 1: Corporate Infrastructure Dominates Extreme Upload Tiers

Advanced filtering via Task 4 proves that channels crossing the 200,000 upload threshold are exclusively tied to global corporate news conglomerates and media syndicates rather than independent creators. This reveals a clear operational divergence between high-frequency content distribution networks and creative personality-driven models.

Insight 2: The Subscriber Engagement Paradox

Evaluating data from Task 7 and 8 demonstrates that absolute subscriber magnitude does not linearly correlate with true audience retention. Niche category clusters (e.g., highly specialized education or instructional verticals) maintain a Viewed by Subscriber coefficient up to three times greater than entertainment domains, indicating a highly dedicated, conversion-ready audience.

7. Portfolio Deployment Strategy

To maximize marketability to technical recruiters, this Power BI Capstone project has been prepared as an open-source case study hosted via GitHub. The repository includes the complete data pipeline lifecycle documentation, clean screenshots of the high-contrast dashboard pages, the localized .pbix project binaries, and this verified analytical report.